Starfish hoodie

size 128 · wide fit, Sweat / French Terry (25 % stretch)

5 parts · 16 steps · medium · about 2.5 h



Front view



Back view

Sewn with: Sewing machine



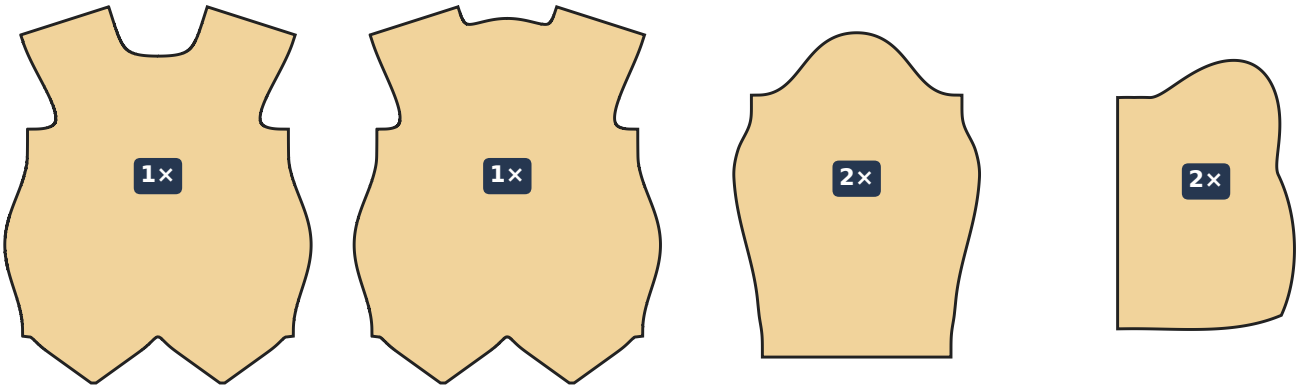
Sewing machine



Iron

Read the steps once before you start. Every picture is drawn from your pattern pieces. Sew a test garment from cheap fabric first and adjust before you cut the good fabric.

Parts



A · Front (cut on fold) B · Back (cut on fold) C · Sleeve E · Hood



D · Sleeve cuff

Seam allowance (included): 1.0 cm · turned-up hems 2.5 cm

Material

Main fabric
1.05 m at 150 cm width

Rib knit
0.15 m at 70 cm width

Sewing machine
narrow zigzag 1 × 2.5 mm (or overlocker); twin needle 4 mm, stitch 3 mm; wide zigzag 4 × 1.5 mm over the edge

Iron
steam

Polyester sewing thread
In the colour of the fabric.

Clear elastic or twill tape
Sewn into the shoulder seams, so they do not stretch out.

Pins or clips
Pin across the seam line.

Chalk or a fabric pen
For notches, labels and placement lines.

Colours in this booklet

Main fabric
right side · wrong side

Rib knit
right side · wrong side

Sew now (machine)

Sew by hand

Already sewn

Fold / press line

Marking (chalk)

Pin

Notch: match them

Right/wrong side of the fabric = the side of the fabric. Left/right (as worn) = seen from the child wearing it: the left sleeve is on the right when you look at the child.

Basics: Sewing machine

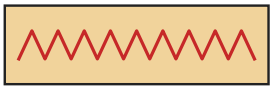
Steps are written for a sewing machine with zigzag. Have an overlocker or a coverstitch machine? Tell the generator which machines you own and the steps use them.

Basics: the sewing machine



Straight stitch

Seams in woven fabric, 2.5 mm; backstitch at the start and the end.



Zigzag

Knit seams (narrow) and finishing the edges (wide).

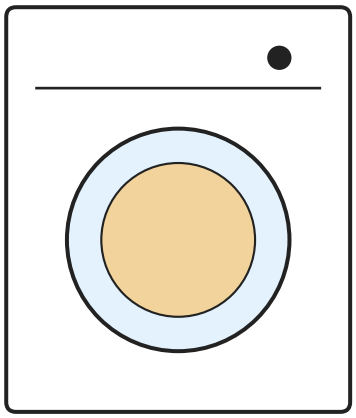


Topstitching

Right side up, with the same distance to the edge all the way; use the edge of the presser foot as a guide.

1 Pre-wash the fabric

1. Wash and dry it the way you will wash the garment later, so it shrinks now, not after sewing.



30–40°

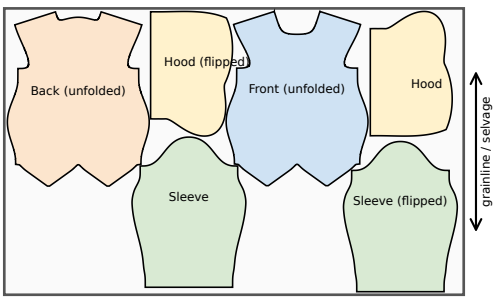
2 Cut out all pieces

Scissors

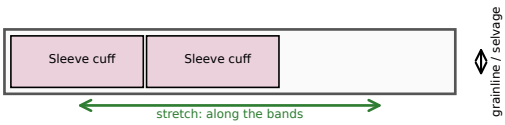
1. Follow the cutting layout
2. Grainline arrows parallel to the selvage.
3. Rib knit: the ribs run along the grainline, so every band stretches along its length (across the fabric).

Tip: Single layer: cut the second piece of a pair with the paper piece turned over (printed side down).

Main fabric · 150 cm · single layer



Rib knit · 70 cm · single layer

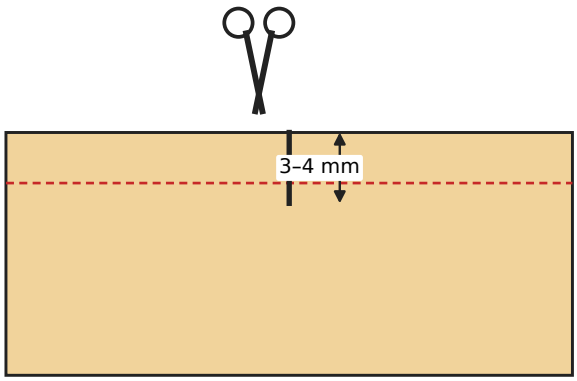


3 Snip the notches

Chalk

1. Cut 3–4 mm into the seam allowance at every notch.

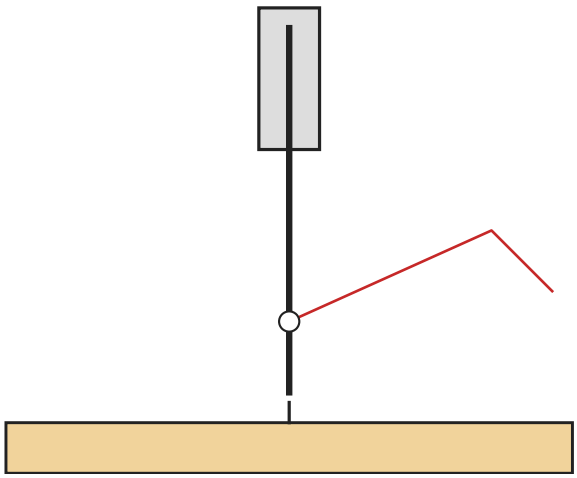
⚠ Important: Never deeper than half the seam allowance.



4 Set up the machine

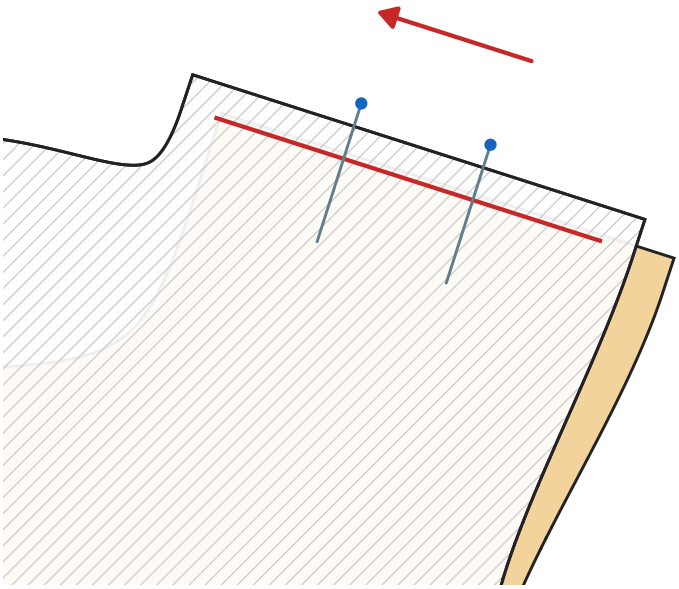
1. Sewing machine: jersey/stretch needle 75–80, polyester thread.
2. Sew a test seam on two layers of a scrap.

Tip: See the basics page at the front of the booklet.



5 Sew the shoulder seams

Sewing machine · Pins



1. Lay front and back right sides together, shoulder edges level, and pin.
2. Sew with 10 mm seam allowance from the neckline to the armhole.
3. Press the allowances towards the back.

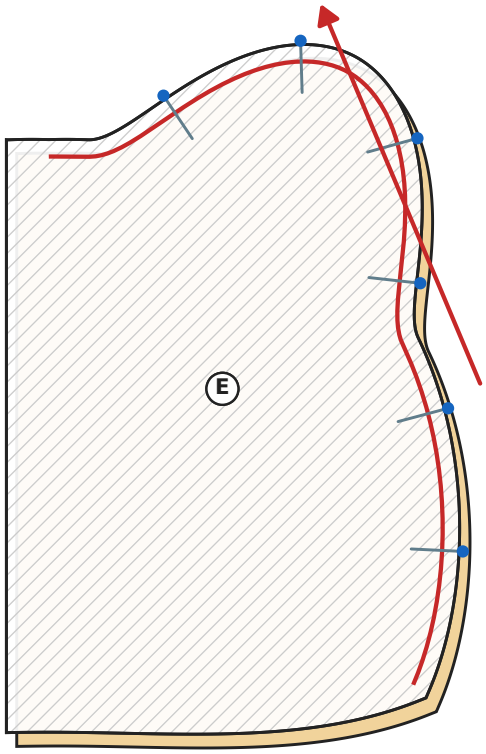
Sewing machine: narrow zigzag 1 × 2.5 mm (or overlocker)

Tip: Knit stretches at the shoulder: sew a strip of clear elastic or twill tape into the seam.

2×

6 Sew the hood centre seam

Sewing machine · Pins



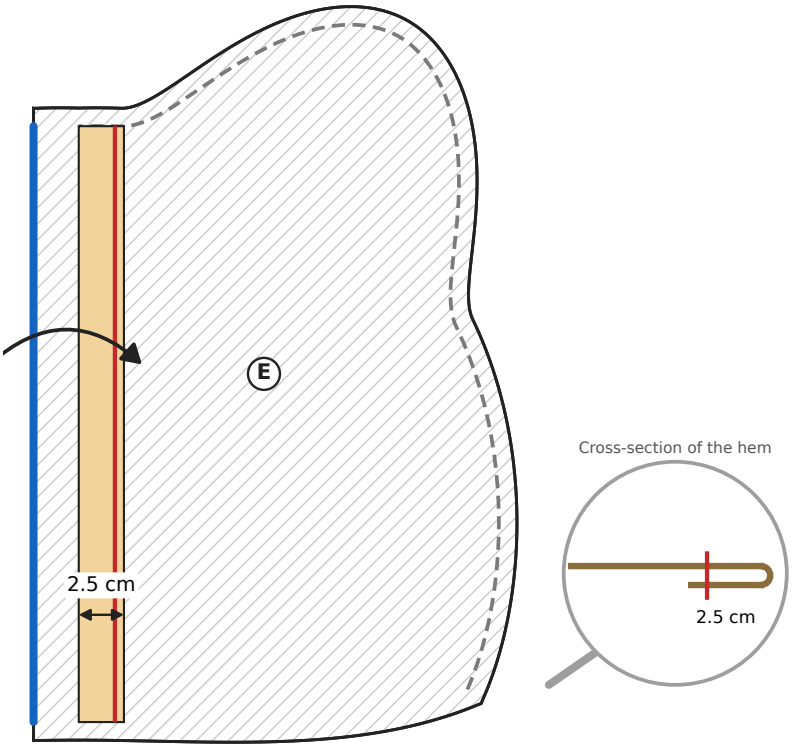
1. Lay the two hood pieces right sides together, curved back edges level, and pin from the neck edge upwards.
2. Sew with 10 mm seam allowance.
3. Press the allowances to one side.

Sewing machine: narrow zigzag 1 × 2.5 mm (or overlocker)

Tip: Topstitch 6 mm beside the seam through the allowances: the seam stays flat.

7 Hem the hood front edge

Sewing machine



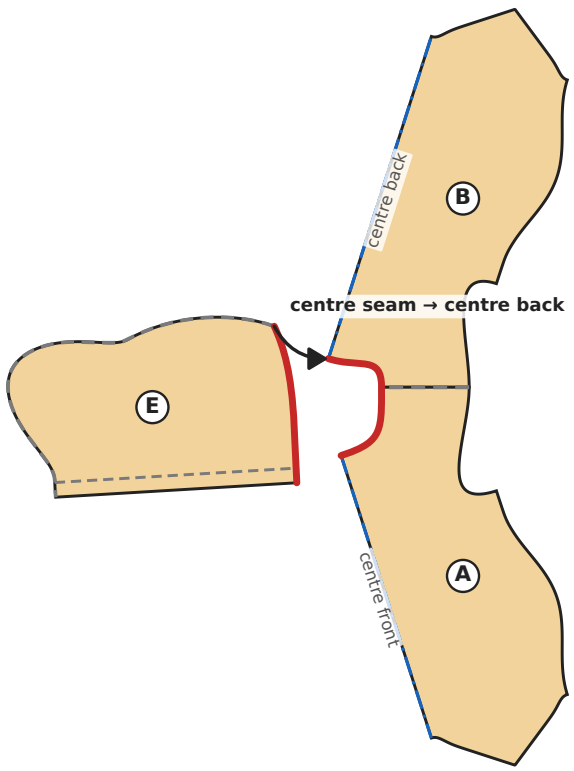
1. Fold the front edge 2.5 cm to the wrong side along the fold line and press.
2. Topstitch 2 cm from the fold, close to the raw edge
3. Leave the ends open for a drawstring.

Sewing machine: twin needle 4 mm, stitch 3 mm

Alternative: Lined hood (option hood_lining): no hem, the lining finishes the front edge and the hood is warmer.

8 Sew the hood on

Sewing machine · Pins

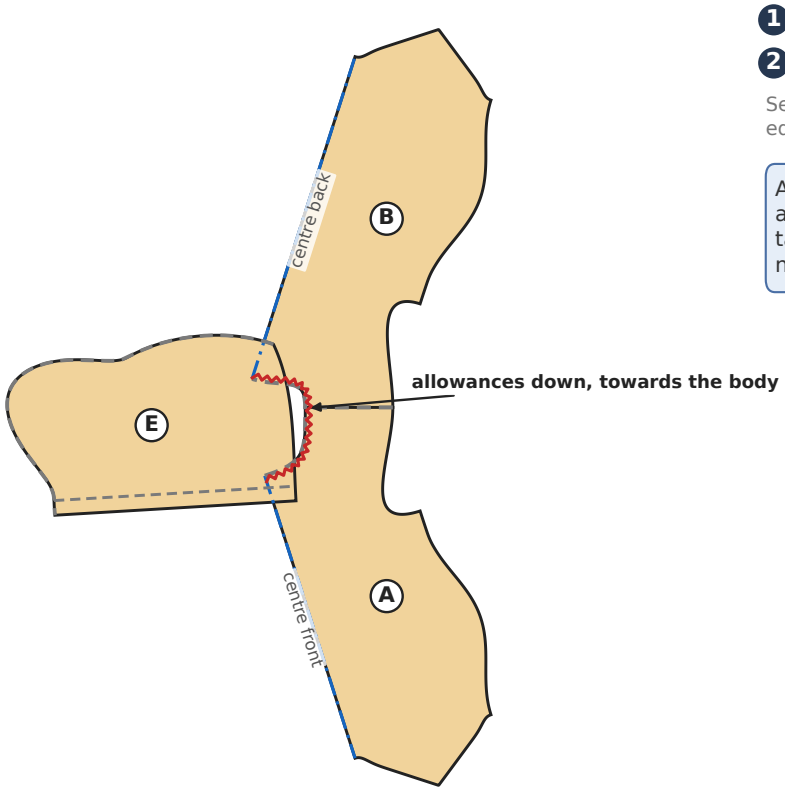


1. Body opened out flat, right side up.
2. Lay the hood right side down onto the neckline, neck edges level.
3. Hood centre seam on centre back, hood notches on the shoulder seams
4. Front edges meet at centre front (overlap 1 cm).
5. Sew all round with 10 mm seam allowance.

Sewing machine: narrow zigzag 1 × 2.5 mm (or overlocker)

9 Finish the neck seam

Sewing machine



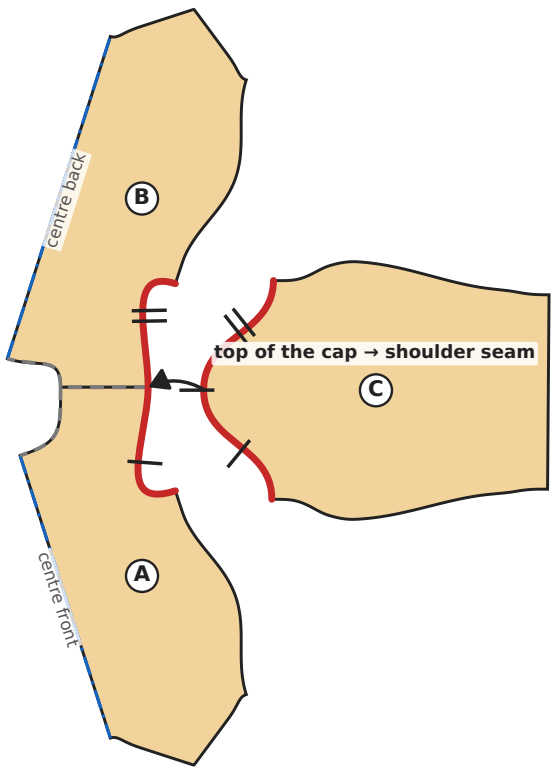
1. Finish both allowances together.
2. Press them down towards the body.

Sewing machine: wide zigzag 4 × 1.5 mm over the edge

Alternative: Neatest finish: cover the allowance with a strip of jersey or cotton tape from shoulder to shoulder over the back neck and stitch it down on both edges.

10 Set in the sleeves

Sewing machine · Pins



1. Open the body out flat, right side up.
2. Lay the sleeve right side down on the armhole: top of the cap on the shoulder seam, single notch on the front, double notch on the back.
3. Pin from the middle to both ends, easing in the extra length of the cap.
4. Sew with 10 mm seam allowance, sleeve underneath
5. Press the allowances towards the sleeve.

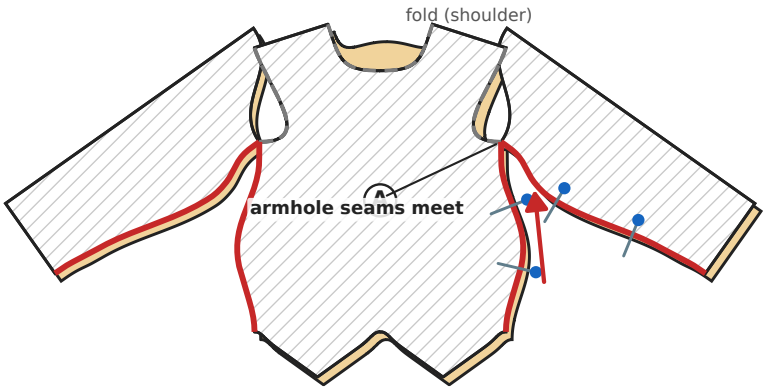
Sewing machine: narrow zigzag 1 × 2.5 mm (or overlocker)

⚠ Important: Sleeve on the bottom: the feed dogs ease the cap in without pleats.

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11 Sew side and sleeve seams

Sewing machine · Pins



1. Fold the garment right sides together at the shoulders, the sleeves folded in half lengthwise.
2. Pin side and sleeve seam so that the armhole seams meet exactly at the underarm.
3. Sew in one go with 10 mm seam allowance, from the hem over the underarm to the sleeve hem.
4. Press the allowances towards the back.

Sewing machine: narrow zigzag 1 × 2.5 mm (or overlocker)

> Alternative: Cuffs first: sew the cuffs on while the sleeve is still flat, then sew side, sleeve and cuff seam in one go. Easier on small sizes.

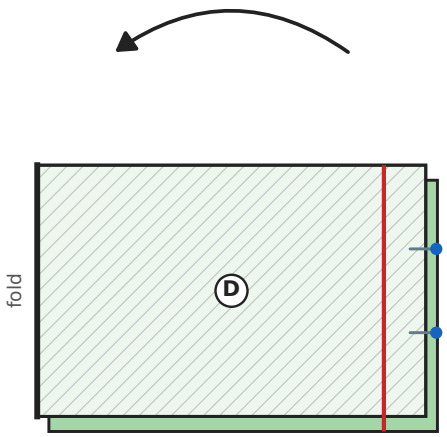
2×

12 Close the cuffs into rings

Sewing machine · Pins

1. Fold each cuff crosswise right sides together, short edges level.
2. Sew with 10 mm seam allowance
3. Finger-press the allowances open.

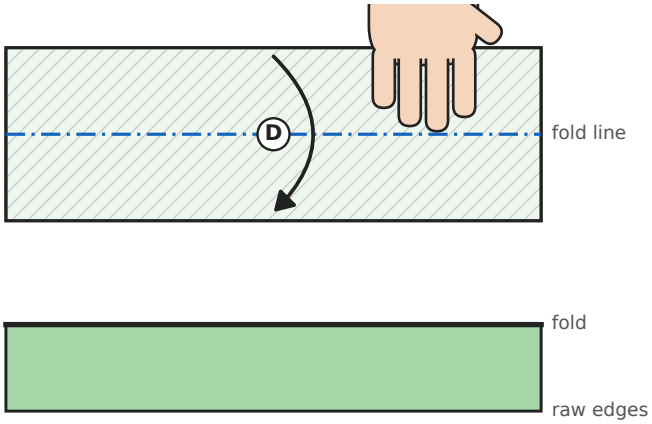
Sewing machine: narrow zigzag 1 × 2.5 mm (or overlocker)



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13 Fold the cuffs in half

Iron



1. Fold the ring lengthwise wrong sides together, raw edges level.
2. The seam allowance is now inside
3. The right side shows all round.

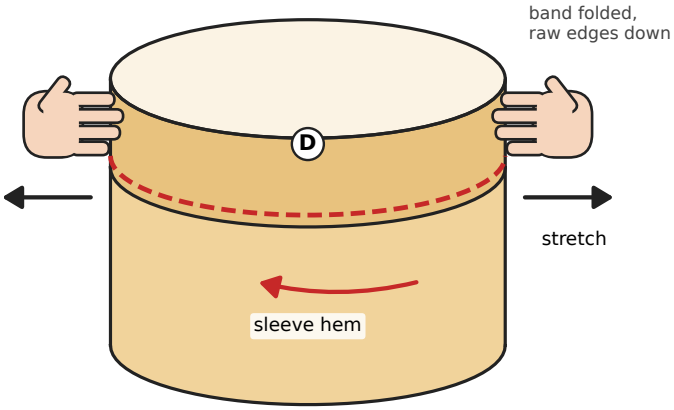
Iron: steam

2×

14 Sew the cuffs on

Sewing machine

Pins



1. Turn the sleeve right side out.
2. Slide the cuff over the sleeve end, raw edges level, cuff seam on the sleeve seam
3. Pin there and at the opposite folds of cuff and sleeve.
4. Sew with the cuff on top, stretching the cuff (not the sleeve) to the length of the sleeve between two pins.

Sewing machine: narrow zigzag 1 × 2.5 mm (or overlocker)

Tip: One hand in front of the presser foot, one behind it: both stretch evenly.

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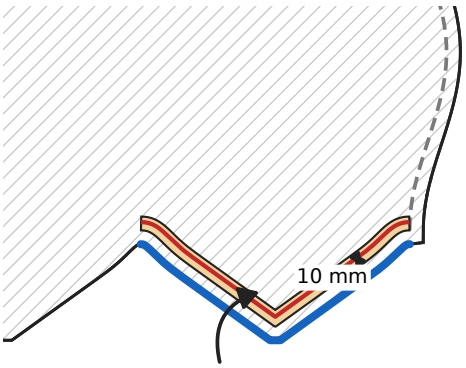
15 Sew the hem

Sewing machine

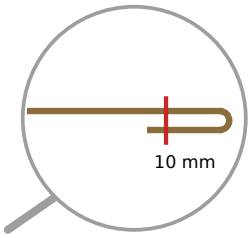
1. Fold the hem 10 mm to the wrong side and press.
2. Sew it down 6 mm from the fold, from the right side.

Sewing machine: twin needle 4 mm, stitch 3 mm

⚠ Important: Shaped hem: clip into every inner point up to 2 mm before the fold and press each point separately.

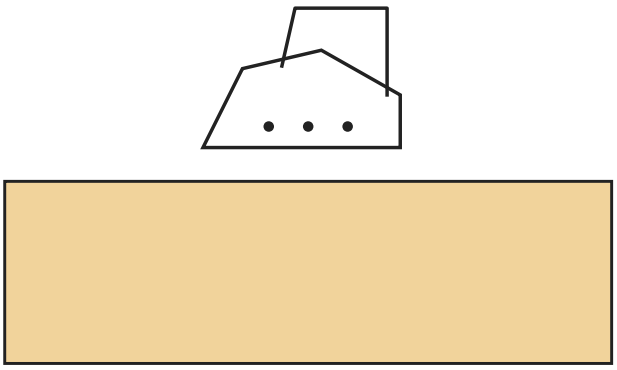


Cross-section of the hem



16 Final pressing

1. Press all seams with steam
2. Do not press down hard on sweat.



Done!



Front view



Back view

After the test garment: Try the test garment on the child: length, chest width, sleeve length and the neckline. Adjust the measurements or options and print the pattern again before you cut the good fabric.

Starfish hoodie · size 128

wide fit, Sweat / French Terry (25 % stretch) · original parametric draft. The pattern is yours to use.

> **Important:** With a new pattern, first sew a **test garment** from inexpensive fabric with similar stretch and check the fit.

Files

- PDF A4 tiles to glue (15 pages + overview): `starfish_hoodie_size128_a4_tiled.pdf`
- Preview A4 tiles: `starfish_hoodie_size128_a4_preview.png`
- SVG master 1:1: `starfish_hoodie_size128_master_1to1.svg`
- Cutting layout: `cutting_layout.png`
- Sewing instructions (PDF): `sewing_instructions.pdf`
- Sewing instructions (Markdown): `sewing_instructions.md`

Printing

- Always print at **100 % / "Actual size"** (not "Fit to page").
- Measure the **5 × 5 cm** control square before you cut.
- Plotter (610 mm roll): the page is 600 mm wide in portrait. In the print dialog choose a custom paper size with exactly this width and length, and turn rotation/scaling off.

Body measurements

Body measurement	Value	Source
Body height	128.0 cm	table
Chest	64.0 cm	table
Waist	57.0 cm	table
Hip	69.0 cm	table
Head	52.5 cm	table
Neck	29.0 cm	table
Inseam	57.5 cm	table
Side length	76.0 cm	table
Arm length	50.0 cm	table
Back length	31.0 cm	table
Shoulder width	10.4 cm	table
Thigh	38.6 cm	table
Ankle	18.6 cm	table

Finished measurements (virtually sewn)

Finished measurement	Value
Chest width	75.0 cm
Hem width	96.5 cm
Length from neck point	47.6 cm
Sleeve length	50.0 cm
Sleeve opening	18.6 cm
Upper-arm width	29.8 cm
Neckline circumference	43.4 cm
Hood height	33.0 cm

Pattern pieces

Piece	Cutting	Fabric	Seam allowance
Front	cut 1 on fold	Main fabric	seam allowance 1.0 cm included
Back	cut 1 on fold	Main fabric	seam allowance 1.0 cm included

Piece	Cutting	Fabric	Seam allowance
Sleeve	cut 2 (mirrored pair)	Main fabric	seam allowance 1.0 cm included
Sleeve cuff	cut 2	Rib knit	seam allowance 1.0 cm included
Hood	cut 2 (mirrored pair)	Main fabric	seam allowance 1.0 cm included; Front edge (casing/facing): 2.5 cm

Fabric needed

- **Main fabric:** about 1.05 m at 150 cm width (single layer, incl. 10 % for shrinkage; folded would need 1.30 m)
- **Rib knit:** about 0.15 m at 70 cm width (single layer, incl. 10 % for shrinkage)
- Design: option hem_band = False (instead of True): needed for the shape.
- Design: option kangaroo = False (instead of True): needed for the shape.
- Creative design 'Starfish': the shape deviates from the base block on purpose; see 'Design' in the validation report.
- Shaped hem: finish it with a narrow hem (overlock or zigzag the edge, turn 1 cm, topstitch) or a facing; clip into every inner point.

Cutting layout: see `cutting_layout.png` (grainline arrow parallel to the selvage, "flipped" = place the piece face down, "×2" = cut through both layers of the folded fabric, which gives a mirrored pair, "(unfolded)" = cut-on-fold piece shown opened out).

Cut on fold (Front, Back): the paper piece is only one half. Its fold edge (dash-dot line, bracket arrows "PLACE ON FOLD") is the centre of the piece: it gets no seam allowance and is never cut. Where the cutting layout shows a piece "(unfolded)" on a single layer, fold the fabric along the grain just wide enough for the piece and place the fold edge on that fold, or trace the paper piece, flip it over along the fold edge and trace the second half.

Validation report

Sewability check

Area	Check	Target	Actual	Δ	Result	Note
Seam pairs	Side seam front ↔ back	30.3 cm	30.3 cm	+0.0 mm	✓ OK	
Seam pairs	Armhole ↔ sleeve cap (ease)	ease 0–8 mm	36.5 cm ↔ 36.8 cm	+3.2 mm	✓ OK	Sweat / French Terry: 0–8 mm
Seam pairs	Shoulder seam front ↔ back	11.9 cm	11.9 cm	+0.0 mm	✓ OK	
Seam pairs	Sleeve seam (front ↔ back)	38.5 cm	38.5 cm	+0.0 mm	✓ OK	
Seam pairs	Sleeve cuff ↔ sleeve hem (stretched)	70–85 % of 26.5 cm	18.6 cm = 70 %	-79.5 mm	✓ OK	rib band is stretched while sewing
Seam pairs	Hood ↔ neckline (each side)	21.7 cm	21.7 cm	-0.0 mm	✓ OK	
Seam pairs	Hood centre seam (left ↔ right)	57.0 cm	57.0 cm	+0.0 mm	✓ OK	
Angles/transitions	Underarm: side seam ↔ armhole front	90°	90.4°	+0.4°	✓ OK	
Angles/transitions	Underarm: side seam ↔ armhole back	90°	90.5°	+0.5°	✓ OK	
Angles/transitions	Shoulder ↔ armhole (front + back)	180°	179.5°	-0.5°	✓ OK	
Angles/transitions	Sleeve cap top (smooth)	180°	178.9°	-1.1°	✓ OK	
Angles/transitions	Sleeve underarm ↔ cap front	90°	90.5°	+0.5°	✓ OK	

Area	Check	Target	Actual	Δ	Result	Note
Angles/transitions	Shoulder ↔ neckline (front + back)	180°	180.5°	+0.5°	✓ OK	
Angles/transitions	Neckline at centre front	90°	90.1°	+0.1°	✓ OK	
Angles/transitions	Neckline at centre back	90°	89.6°	-0.4°	✓ OK	
Angles/transitions	Hem at centre front	92°	92.2°	+0.0°	✓ OK	changed by the design (intended)
Angles/transitions	Hem ↔ side seam front	93°	92.6°	-0.0°	✓ OK	changed by the design (intended)
Angles/transitions	Hood: front edge ↔ neck seam	90°	90.0°	+0.0°	✓ OK	
Measurements	Chest circumference pattern = body + ease	74.9 cm	75.0 cm	+0.1 %	✓ OK	body 64.0 cm + +17 % ease
Measurements	Neckline (fabric edge) fits over the head	≥ 52.5 cm	54.3 cm	+18.0 mm	✓ OK	neckline incl. stretch ≥ head circumference
Measurements	Hand fits through the sleeve cuff	≥ 19.8 cm	33.4 cm	+135.7 mm	✓ OK	hand circumference ≈ 1.55 × wrist
Common practice	Chest ease for fit 'wide'	14 – 28 %	17 %	+0 %	✓ OK	guide value for this fit
Common practice	Sleeve cap height for this shoulder	5.2 cm – 9.9 cm	9.4 cm	+0.0 mm	✓ OK	lower cap for a dropped shoulder
Measurements	Sleeve length plausible	45.0 cm – 62.5 cm	50.0 cm	+0.0 mm	✓ OK	arm length 50.0 cm
Measurements	Body length plausible	38.4 cm – 79.4 cm	47.6 cm	+0.0 mm	✓ OK	from the neck point
Measurements	Hood tall enough	≥ 28.9 cm	33.0 cm	+40.8 mm	✓ OK	hood height ≥ 0.55 × head circumference
Geometry	Front: outline & seam allowance	closed, SA present	ok		✓ OK	seam allowance 1.0 cm included
Labels	Front: grainline & cutting	present	cut 1 on fold		✓ OK	Main fabric
Geometry	Back: outline & seam allowance	closed, SA present	ok		✓ OK	seam allowance 1.0 cm included
Labels	Back: grainline & cutting	present	cut 1 on fold		✓ OK	Main fabric
Geometry	Sleeve: outline & seam allowance	closed, SA present	ok		✓ OK	seam allowance 1.0 cm included
Labels	Sleeve: grainline & cutting	present	cut 2 (mirrored pair)		✓ OK	Main fabric
Geometry	Sleeve cuff: outline & seam allowance	closed, SA present	ok		✓ OK	seam allowance 1.0 cm included
Labels	Sleeve cuff: grainline & cutting	present	cut 2		✓ OK	Rib knit
Geometry	Hood: outline & seam allowance	closed, SA present	ok		✓ OK	seam allowance 1.0 cm included; Front edge (casing/facing): 2.5 cm
Labels	Hood: grainline & cutting	present	cut 2 (mirrored pair)		✓ OK	Main fabric
Geometry	Hood: turned allowance lies flat	≤ 0.5 mm	0.0 mm		✓ OK	Front edge (casing/facing): corners mirrored at the fold line

Area	Check	Target	Actual	Δ	Result	Note
Geometry	Front: cut-on-fold edge	straight, no SA, 90° corners, on grain	ok, corners ≤ 0.1° off 90°		✓ OK	Centre (fold)
Geometry	Back: cut-on-fold edge	straight, no SA, 90° corners, on grain	ok, corners ≤ 0.4° off 90°		✓ OK	Centre (fold)
Labels	Front: notches	on the cutting line	1 notch mark(s)		✓ OK	
Labels	Back: notches	on the cutting line	2 notch mark(s)		✓ OK	
Labels	Sleeve: notches	on the cutting line	4 notch mark(s)		✓ OK	1 at corners
Design (intended)	Side seam: bulge +3.0 cm at 55 %	base block	+0.0 … +3.0 cm		✓ OK	intended deviation (design 'Starfish')
Design (intended)	Sleeve seam: profile +3.0 cm at 30 %, +0.5 cm at 85 %	base block	+0.0 … +3.0 cm		✓ OK	intended deviation (design 'Starfish')
Design (intended)	Hood centre seam: peak +9.0 cm at 70 %	base block	+0.0 … +9.0 cm		✓ OK	intended deviation (design 'Starfish')
Design (intended)	Hem: 2 × points, 7.0 cm deep	base block	+0.0 … +7.0 cm		✓ OK	intended deviation (design 'Starfish')
Freedom of movement	Side seam: girth after the design	≥ 60.8 cm	75.0 cm	+0.1 cm vs. base	✓ OK	not tighter than the base block (at 8 %)
Freedom of movement	Sleeve seam: girth after the design	≥ 16.4 cm	26.6 cm	+0.0 cm vs. base	✓ OK	not tighter than the base block (at 95 %)
Freedom of movement	Hem: girth after the design	≥ 60.0 cm	96.5 cm	+18.6 cm vs. base	✓ OK	not tighter than the base block (at 100 %)
Design sewability	Side seam: steepness of the shape	≤ 1.0	0.36		✓ OK	offset change per length along the seam (1.0 = 45°)
Design sewability	Sleeve seam: steepness of the shape	≤ 1.0	0.58		✓ OK	offset change per length along the seam (1.0 = 45°)
Design sewability	Hem: points can be sewn and turned	≥ 60°	120°		✓ OK	outer points of the reshaped edge
Design sewability	Hem: inner corners	≥ 60°	none		✓ OK	

Result: 53 OK, 0 warning(s), 0 error(s): the pattern is sewable.

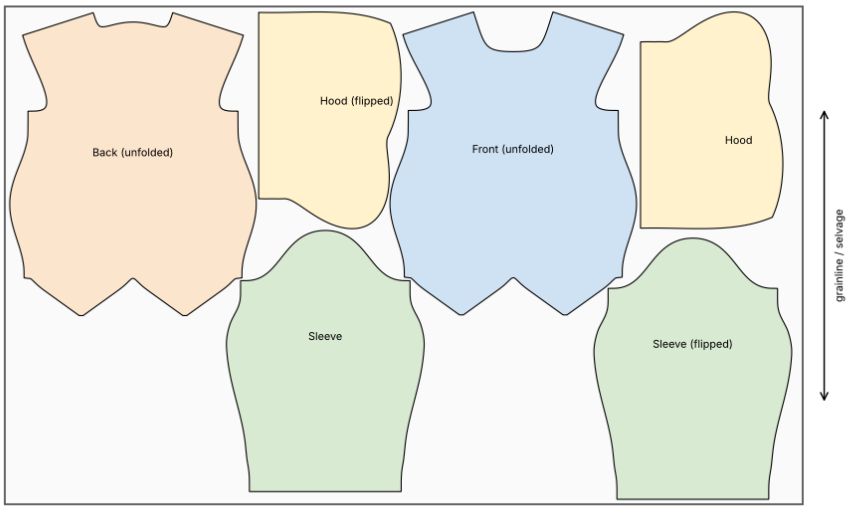
Construction values

Override any of these with the option `construction`, values in mm (e.g. `{"construction": {"chest_width": 749}}`).

Name	Value	Derived from	Note
`chest_width`	74.9 cm	chest × (1 + 17 % ease)	
`neck_width_half`	8.1 cm	neck / 5 + 1.5 cm	
`front_neck_depth`	7.2 cm	0.95 × neck_width_half − 0.5 cm	
`back_neck_depth`	1.5 cm	fixed	
`shoulder_length`	11.9 cm	shoulder + allowance for fit 'wide'	
`shoulder_slope`	17	fixed	
`armhole_depth`	18.2 cm	18 % of chest_width + 2.5 cm + 3 %	
`body_length`	47.6 cm	back length + 13 % of height	

Cutting layout · Starfish hoodie size 128

Main fabric: 150 cm wide, single layer
needed ≈ 1.05 m



Rib knit: 70 cm wide, single layer
needed ≈ 0.15 m

